# Referee package (DRAFT) – SC-SCOPE: the second- and third-cumulant endpoint closure of the selection floor

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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## 1. Purpose and scope

This referee document concerns the SC-SCOPE programme: the second- and third-cumulant scope of the selection floor and its endpoint closure (sunset and quartic-difference channels). **Scope:** the all-orders endpoint at $I = 2 \times 10^{-3}$. **Not claimed:** the closure is thin (sunset-limited, near-critical); it is a near-critical selection boundary, not a comfortable margin.

## 2. The result

The SC-SCOPE third-cumulant joint inequality is positive at the endpoint: each channel (sunset cap $\times 1.13$, realized quartic $R_{\max} = 0.385 < 0.634$ with the Parseval-pinned $1/(4\pi^2)$ convolution measure) and their joint with the second-order cost satisfy joint $> 1$. The endpoint sits just below the structural sunset cap; the selection floor closes (thin).

## 3. Structure of the argument

The realized quartic difference is computed directly ($R_{\max} \approx 0.385$) and certified $< 0.634$ once the convolution normalisation $1/(4\pi^2)$ is DERIVED from the $(2\pi)^3$ measure (Parseval ratio 1.0000). The sunset channel saturates at $\times 1.13$. Assembling the second- and third-order costs against the layer margin gives joint $> 1$ at the endpoint, with the thinness shown structural (sunset-limited) and sign-stable across nearby $(I, \mu^2)$.

## 4. Numerical reproduction

Five scripts reproduce the per-transfer quartic, the floor sharpening, the joint pairing, the M-endpoint evaluation and the quartic normalisation certificate; each carries asserts. Run them from the bundle.

## 5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

($\alpha$) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.

($\beta$) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.

($\gamma$) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

## 6. Result footer

```
Result ID:               B5-BEYOND-LAYER-BOUND / SC-SCOPE referee
                         package (DRAFT)
Precise statement:       At the all-orders endpoint I=2e-3 the SC-SCOPE
                         third-cumulant joint > 1: realized quartic
                         R_max=0.385<0.634 (with the Parseval-pinned
                         1/(4pi^2) convolution measure, ratio 1.0000),
                         sunset cap x1.13, joint of second+third-order
                         costs against the layer margin positive (thin,
                         sunset-limited, near-critical, sign-stable
                         nearby).
Scope:                   SC-SCOPE all-orders endpoint selection floor.
                         Thin (sunset-limited near-critical).
Evidence grade:          third-cumulant endpoint closure,
                         thin-certified (T5); EXECUTED (5 scripts PASS).
Reproduction command:    python
                         codes/vacuum/scscope_ghat4_pertransfer.py ;
                         scscope_quartic_certificate.py ;
                         scscope_joint_pairing.py ; ...
Falsifier:               a recomputed joint <=1 at the endpoint; or a
                         convolution-normalisation that doubles Ghat4
                         (R_max>0.634).
Tier before / after:     DRAFT referee package; SC-SCOPE endpoint
                         thin-certified T5. NOT operator-confirmed.
No-overclaim statement:  the endpoint closes THIN (sunset-limited
                         near-critical); not a comfortable
                         all-intensity statement.
Next required action:    operator review + confirm -> PUBLISHED bundle.
```